No-Regret is Not No-Harm:
Bandits with Consumable Action Sets

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Abstract

Sublinear regret is the standard justification for deploying sequential decision-making systems. We show that this justification is vacuous when actions consume the action set. We study bandits in which pulling an arm may destroy it, and in which each destroyed arm imposes a permanent negative externality on every subsequent reward. We prove a two-sided separation: a policy achieving *exactly zero* regret against the best-available-arm benchmark can incur value loss $m\delta E$, unbounded in both the pool size m and the externality magnitude δE, and can drive total value negative while the optimal policy remains positive; conversely, any policy near the system optimum is *forced* to report private regret $m\epsilon$. The metric does not merely fail to separate good from destructive policies—it orders them backwards. The mechanism is that regret is measured against an optimum the learner has itself degraded. We further establish that the parameter governing optimal play, the expected marginal externality $\kappa_a = p_a e_a$, is (i) exactly the quantity that determines whether greedy is optimal, and (ii) subject to an irreducible estimation error $\mathrm{SE} \geq 2\sigma/\sqrt{\min(T, \sum_j 1/p_j)}$ that *does not decrease with the horizon*, because each arm supplies exactly one before/after transition and the sample budget is capped by consumption rather than time. The quantity that matters most cannot be estimated arbitrarily accurately, and its attainable precision degrades exactly as it becomes more important. We give a closed-form correction (ECI, exact under constant κ, with a proven though loose bound on its general approximation gap), characterise an exactly-solvable special case, and show that sequential multi-agent consumption on a shared pool reduces exactly to the single-agent case, so no price-of-anarchy separately arises there. We further stress-test the independence assumption underlying every result: under correlated, cluster-level destruction the core characterisation appears to survive exactly (verified, not proven) while the correction requires an explicit adjustment, which we derive and validate against exact and scaled ground truth. We release an open-source testbed, Attrition, with a regression test for every claim.

1 Introduction

Consider an orchestrator routing tasks across API-backed tools. Some tools are robust; others trip a rate limit, exhaust a quota, or get a key revoked under heavy use, disappearing for the remainder of the session. Tools are not independent: those sharing upstream infrastructure—a vendor, an IP pool, a quota tier—degrade together when one is burned.

Table 1 shows two routers on identical instances. The first has a flawless record: at every step it pulls the highest-value available tool, so its regret against the best-available benchmark is identically zero. It ends on a fallback tool worth 0.470 per call with two tools remaining. The second reports regret of 0.25 per step—and is sitting at 0.782 per call, 66% higher, with four tools remaining.

The greedy router's zero-regret record is not evidence of good behaviour. It is an artefact of a benchmark that degrades in lockstep with the damage being done: every call greedy makes is optimal *given the ecosystem it has already ruined.*

Figure 1 shows the phenomenon in its cleanest form. As the coupling between arms strengthens, the greedy policy's cumulative regret stays pinned at exactly zero while its realised value falls monotonically and eventually turns negative—below the value of taking no action at all—even as the optimal policy remains comfortably positive. The two panels are computed on the same instances. A practitioner monitoring the left panel would see nothing wrong at any point.

This paper formalises that failure. Our contributions:

Table 1: Two routers, identical seed and tool set, single session. "realised" is the reward actually obtained, $v_{a_t} - B(S_t)$; "regret" is the per-step quantity $\max_{a \in S_t}(v_a - B(S_t)) - (v_{a_t} - B(S_t))$, i.e. instantaneous regret against the best arm available at that step, not a cumulative or expected quantity. The policy that looks worse by the standard metric achieves the better outcome.

router	step	realised	regret	tools left
greedy	0	1.150	0.0000	6
greedy	2	0.844	0.0000	4
greedy	7–13	0.470	**0.0000**	**2**
ECI	0	1.150	0.0000	6
ECI	1–4	0.988	0.0500	5
ECI	5–13	**0.782**	0.2500	**4**

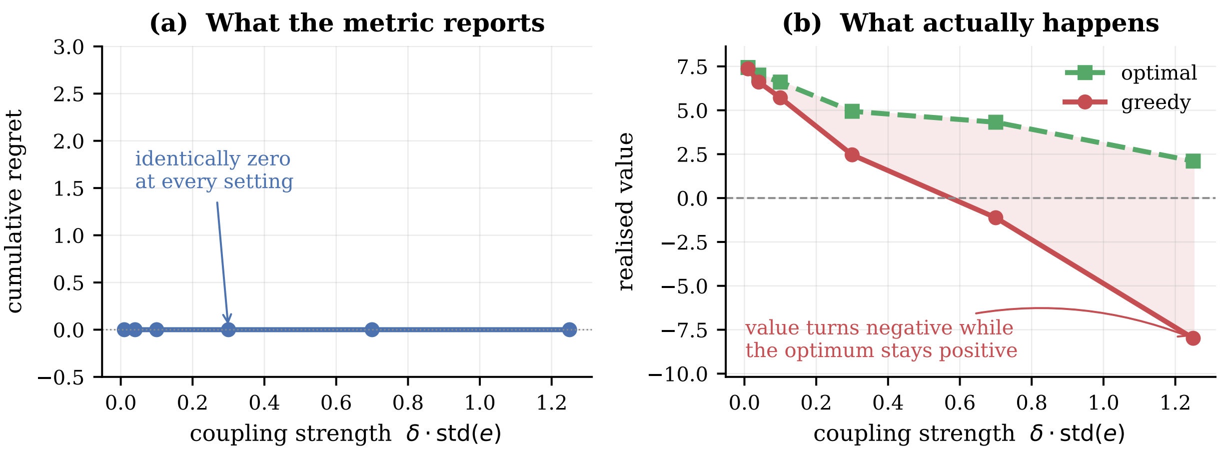

Figure 1: The central phenomenon. **(a)** Greedy's cumulative regret against the best-available-arm benchmark, computed exactly: identically zero at every coupling strength. This is true by construction—greedy maximises the benchmark quantity at each step—and is shown not as a surprise in itself but as the reference against which panel (b) should be read. **(b)** Realised value on the same instances. The shaded region is the loss. At strong coupling the zero-regret policy finishes net-negative while the optimum stays positive. Averages over 20 instances, $n = 6$, $T = 10$, exact dynamic programming.

- **Four domain instantiations**—agent tool ecosystems, adaptive cancer therapy under competitive release, platform trials, and CMC design-space development—each simulated, with fit quality stated honestly (Section 10).
- **Theorem 1** (main result). A zero-regret policy can incur value loss $m\delta E$, unbounded in pool size and externality magnitude, and can finish net-negative while the optimum is positive. No-regret is not no-harm.
- **Theorem 2** (converse). Any near-system-optimal policy must incur private regret $m\epsilon$. Preserving value *requires* looking bad on the standard metric, and by Corollary 3 the amount it looks bad by is proportional to the value it delivers.
- **Theorem 4**. Greedy is optimal if and only if $\kappa_a = p_a e_a$ is constant across arms. Neither factor alone is the invariant.
- **Theorem 5**. The externality cannot be estimated to arbitrary precision: the error floor does not decrease with the horizon, because the data-generating process is the consumption itself.

- **Theorem 12**. In the pure-sequencing regime the optimal order is ascending in e; arm values are irrelevant.
- A closed-form correction (ECI), exact under constant κ (Proposition 10) with a proven general bound (Theorem 9), and a rollout policy reaching 99.5–99.9% of optimum.
- **A sequential-equivalence theorem** (long-form edition, Section 11): multi-agent consumption on a shared pool reduces exactly to a single-agent problem over a longer horizon—so no price of anarchy arises from consumption externalities alone, and the target for mechanism design is free-riding specifically, not coordination.
- A stress test of the independence assumption (Section 12): under correlated cluster-level destruction, the characterisation appears to survive exactly (a verified conjecture) while the correction fails and is repaired (SECI), validated against exact ground truth at up to twice the originally checked scale.
- An open-source testbed, Attrition, with 50 experiments and 76 regression tests—at least one per theorem, several per result with multiple independently checkable parts.

2 Related work

Rotting bandits [8, 11, 12] model rewards that decay with the number of pulls (*rested*) or with elapsed time (*restless*); Seznec et al. [12] give a single adaptive-window policy near-optimal for both. Decay there is gradual rather than removal, and critically the learner is assumed to know which regime it is in. **Mortal bandits** [2] have arms with stochastic lifetimes, but expiry is exogenous. **Blocking bandits** make a pulled arm temporarily unavailable. **Sleeping bandits** [6] allow arbitrary exogenous availability. None of these couple arms: destroying one does not change another's reward distribution.

Bandits with knapsacks [1] is the closest structural neighbour: playing an arm consumes resources and the process halts when a budget is exhausted. The budget is global and exogenous, and consumption leaves other arms' rewards untouched. Here consumption permanently degrades *every* future reward. That difference is why the governing quantity is $\kappa = p \cdot e$ rather than a consumption rate.

Bandit learning with positive externalities [13] is the only prior work we are aware of that places externalities in a bandit model. There, satisfied users attract similar users, so externalities are positive and act on the *arrival process*; the authors show UCB incurs linear regret and develop balanced exploration. Our externalities are negative, act on *rewards*, and are triggered by irreversible destruction. The parallel is worth drawing: in both settings a standard algorithm fails badly, indicating that externality structure of either sign breaks classical guarantees.

Estimation under adaptive data collection. A separate literature studies bias and variance in estimates built from bandit-collected data: Nie et al. [10] show common bandit algorithms produce systematically biased sample means because the data-generating process is entangled with the decisions being evaluated. Theorem 5 is in this family but is not a bias result: we hold the estimator unbiased (ordinary least squares) and instead bound its *variance*, and the mechanism is specific to this setting—each arm supplies exactly one before/after transition because destruction is permanent, capping the information at the pool size regardless of how long data collection continues. Adaptive-collection bias and consumption-capped variance are complementary difficulties that would compound in a system exhibiting both.

Cascading and interacting bandits couple arms within a round—the reward of one depends on which others were displayed or examined—but the coupling is simultaneous and leaves the action set intact across rounds. **Combinatorial bandits** select subsets with structured rewards, again without destroying arms. **Safe and conservative bandits** constrain reward or maintain a baseline while the arm remains available; the constraint restricts *which* actions may be taken, not what taking them removes. **Reinforcement learning with irreversible actions and safe exploration** studies states from which return is impossible, and typically seeks to *avoid* irreversibility rather than to price it and schedule it optimally. **Open multi-agent bandits** model a population of learners that arrives and departs exogenously; the churn is in the decision makers, not the actions, and is independent of what is played.

Table 2: Notation.

symbol	meaning
v_a	mean reward of arm a
p_a	probability that pulling a destroys it
e_a	externality coefficient of a
δ	externality scale
$\kappa_a = p_a e_a$	expected marginal externality of one pull of a
S_t	set of arms available at round t
$B(S) = \delta \sum_{i \notin S} e_i$	accumulated burden
$N_{\text{exh}} = \sum_j G_j$, $G_j \sim \text{Geom}(p_j)$	pulls until the pool is exhausted
$N = \min(T, N_{\text{exh}})$	realised episode length
$W(\pi) = \mathbb{E}_\pi[\sum_t r_t]$	system value of policy π
$V^\star = \max_\pi W(\pi)$	optimal system value
$R_{\text{sys}}(\pi) = V^\star - W(\pi)$	system-value regret
$I(a,t) = v_a - \delta\kappa_a(T-t)$	externality-corrected index (ECI)

The novelty claim, stated precisely. We do not claim novelty for any one ingredient. Irreversible arm removal appears in mortal bandits; consumption appears in knapsacks; decay with pulls appears in rotting bandits; cross-arm externalities appear in [13]. The combination we study—*(a)* removal that is permanent, *(b)* triggered by the learner's own action, *(c)* carrying a negative externality onto every remaining arm's reward, and *(d)* evaluated against a benchmark that conditions on the already-damaged state—is, to our knowledge, not covered by any of them. It is property (d) combined with (b) that produces Theorem 1: the benchmark and the damage move together. We invite correction on this point and treat it as the claim most in need of scrutiny.

3 Setting

Arms $1, \ldots, n$. Arm a has mean reward v_a, destruction probability $p_a \in (0,1]$, and externality coefficient $e_a \geq 0$. Let S_t be the set of available arms at round t and define the accumulated burden

$$B(S) \;=\; \delta \sum_{i \notin S} e_i.$$

Pulling $a \in S_t$ yields reward $v_a - B(S_t) + \eta_t$ with η_t sub-Gaussian with parameter σ, and destroys a with probability p_a, permanently adding δe_a to the burden. The horizon is T. The value function is

$$V(S,t) = \max_{a \in S} \Big[v_a - B(S) + p_a V(S \setminus \{a\}, t{+}1) + (1-p_a) V(S, t{+}1) \Big].$$

Define the *expected marginal externality*

$$\boxed{\kappa_a \;=\; p_a\, e_a}$$

the expected permanent burden created by one pull of a.

Benchmark. We measure regret against the best available arm, $\max_{a \in S_t} \big(v_a - B(S_t)\big)$. This is the benchmark deployed systems use, and Theorem 1 is a statement about that practice. Value is measured against the exact optimum $V(S_0, 0)$, computed by dynamic programming over subsets for $n \leq 8$ and by rollout above that.

4 A zero-regret policy can destroy unbounded value

Theorem 1 (Vacuity of no-regret under consumption)**.** *For any $M > 0$ there is an instance on which greedy attains regret exactly 0 against the best-available benchmark while $V^\star - V_{\text{greedy}} \geq M$. Moreover the loss can exceed $V^\star$, making the zero-regret policy net-negative while the optimum is positive.*

Proof. Fix $m \geq 1$, $E > 0$, $0 < \epsilon < \delta E$. Take $n = m+1$ arms: one *hot* arm H with $v_H = 1$, $e_H = E$, and m *safe* arms with $v_S = 1 - \epsilon$, $e_S = 0$. Set $p_a = 1$ for all arms and $T = n$, so each arm is pulled exactly once and only the order is chosen.

Greedy pulls H first since $1 > 1 - \epsilon$. H is destroyed, adding permanent burden δE, and each of the m subsequent pulls yields $(1 - \epsilon) - \delta E$:

$$V_{\text{greedy}} = 1 + m(1 - \epsilon - \delta E).$$

Deferring H to last yields $1 - \epsilon$ on each safe pull with no burden, and 1 on the final pull of H, whose burden is charged only to later rounds, of which there are none:

$$V^\star \geq m(1 - \epsilon) + 1.$$

Subtracting, $V^\star - V_{\text{greedy}} = m\,\delta E$, unbounded in m and in δE. Choosing $\delta E > 1$ gives $V_{\text{greedy}} < 0 < V^\star$.

At every round greedy pulls $\arg\max_{a \in S}(v_a - B(S))$, which is the definition of the benchmark; its instantaneous regret is 0 at every state. □

The closed form matches exact dynamic programming to machine precision at all 15 settings tested, $m \in \{1, 2, 4, 8, 12\}$ and $\delta E \in \{0.5, 1, 2\}$. At $m = 12$, $\delta E = 2$: $V^\star = 12.88$, $V_{\text{greedy}} = -11.12$, gap exactly 24.0000, regret 0.0000.

Scaling. The gap is linear in the pool size m, not in the horizon T. With $p = 1$ the pool is exhausted after n pulls, so T beyond n is irrelevant; under $p < 1$ the gap grows with T only until $T \approx \sum_j 1/p_j$ and then saturates at the same sample-budget ceiling that appears in Theorem 5. Pool size is the operative quantity because greedy's error is paid once per subsequent pull.

Theorem 2 (Converse: optimality forces private regret)**.** *In the family of Theorem 1, the optimal policy incurs cumulative private regret exactly $m\epsilon$ against the best-available-arm benchmark. Hence for any $M > 0$ there are instances on which every policy within ϵm of the system optimum has private regret at least M.*

Proof. The optimal policy defers H to the final round. At each of the m earlier rounds the best available arm is H (value 1 against $1 - \epsilon$), and the policy declines it, incurring instantaneous private regret ϵ. Summing over the m deferrals gives $m\epsilon$. Any policy that instead pulls H early forfeits δE per remaining round, so for $\epsilon < \delta E$ deferral is necessary for near-optimality. Taking $m \to \infty$ gives the second claim. □

Exact dynamic programming confirms the closed form at all eight settings tested, $m \in \{2, 4, 8, 12\}$ and $\delta E \in \{0.5, 1.5\}$: the optimal policy's private regret is $m\epsilon$ to machine precision while its system-value advantage over greedy is $m\delta E$.

Together with Theorem 1 this closes the separation in both directions. A zero-regret policy may destroy unbounded value, and—the sharper statement—a policy that preserves value *must* report regret. The metric does not merely fail to separate the two; it orders them backwards.

Corollary 3 (The reported regret tracks the value delivered)**.** *In the same family the optimal policy's private regret is $m\epsilon$ and its system-value advantage is $m\delta E$, so their ratio is $\epsilon/(\delta E)$, independent of m.*

Sweeping $\epsilon \in \{0.01, 0.05, 0.10, 0.20, 0.40\}$ at $m = 8$, $\delta E = 1$: the system gain is 8.0000 throughout while private regret runs $0.08, 0.40, 0.80, 1.60, 3.20$—ratios of exactly ϵ.

A second benchmark. Because the choice of benchmark determines the verdict, we report both. Define *system-value regret* $R_{\text{sys}}(\pi) = V^\star - W(\pi)$ against the ex-ante optimal policy. On random instances ($n = 6$, $T = 8$, 15 instances, exact DP):

policy	system value	system regret	private regret
optimal	5.4048	0.0000	0.9489
ECI	5.3315	0.0734	0.6332
greedy	4.4061	0.9987	**0.0000**

The two regret columns rank the three policies in opposite orders. The algorithms are fixed; the benchmark alone determines which appears best.

Interpretation: private versus system value. Define the *system value* of a policy π as its expected total reward, $W(\pi) = \mathbb{E}_\pi\big[\sum_{t=1}^{T} r_t\big]$, so that $V^\star = \max_\pi W(\pi)$, and define the *private value* of an action at state S as its own immediate reward $v_a - B(S)$. Greedy maximises private value at every step. The optimal policy additionally accounts for $\delta\kappa_a(T-t)$—the reward removed from *all subsequent actions* by this one. Regret against the best-available benchmark compares private values only, so it is invariant to that term. Under externalities private and system value diverge, and a guarantee stated in the former says nothing about the latter.

5 When is greedy optimal?

Theorem 4 (Characterisation)**.** *Fix the burden model of Section 3 with additively separable burden.*

(i) (Sufficiency.) *If $\kappa_a = \kappa$ for all arms a, then greedy is optimal at* every *state S and* every *horizon T, for every choice of $\{v_a\}$.*

(ii) (Necessity.) *If $\kappa_a \neq \kappa_b$ for some pair a, b, then there exist $\{v_a\}$ and a horizon T for which greedy is strictly suboptimal.*

The asymmetry in the quantifiers is deliberate and necessary. Constant κ gives *universal* optimality; varying κ gives *existence* of a failure instance, not failure on every instance. Varying κ with, say, all v_a equal, or a horizon so short that the relevant arms are never reached, need not produce a strict gap. The one-line summary—"greedy is optimal iff κ is constant"—is correct only when read as universal optimality versus existence of a counterexample.

Proof. Write the total value of a policy π over its realised episode as

$$W(\pi) \;=\; \mathbb{E}_\pi\Big[\textstyle\sum_t v_{a_t}\Big] \;-\; \mathbb{E}_\pi\Big[\textstyle\sum_t B(S_t)\Big],$$

and consider the burden term. A destruction at round s imposes δe_{a_s} on every later round of the episode, so

$$\mathbb{E}_\pi\Big[\textstyle\sum_t B(S_t)\Big] \;=\; \delta\,\mathbb{E}_\pi\Big[\textstyle\sum_s \mathbf{1}[\text{destruction at } s]\; e_{a_s}\,(N-s)\Big],$$

where N is the episode length. Conditioning on the history and the arm chosen at round s, the destruction indicator has mean p_{a_s}, so round s contributes $p_{a_s}e_{a_s} = \kappa_{a_s}$ in expectation. Under the hypothesis $\kappa_a \equiv \kappa$ this is the same constant *whichever arm the policy selects*, giving

$$\mathbb{E}_\pi\Big[\textstyle\sum_t B(S_t)\Big] \;=\; \delta\kappa\,\mathbb{E}\Big[\textstyle\sum_s (N-s)\Big] \;=\; \delta\kappa\,\mathbb{E}\Big[\tfrac{N(N-1)}{2}\Big].$$

It remains to note that N is itself policy-independent. Arm j absorbs exactly $G_j \sim \text{Geometric}(p_j)$ pulls before destruction, and every pull is allocated to exactly one arm, so the number of pulls until the pool is exhausted is $N_{\text{exh}} = \sum_j G_j$ *regardless of the order in which a policy allocates them.* Hence $N = \min(T, N_{\text{exh}})$ has a policy-independent distribution.

The burden term is therefore a constant common to all policies, so

$$\arg\max_\pi W(\pi) \;=\; \arg\max_\pi \mathbb{E}_\pi\Big[\textstyle\sum_t v_{a_t}\Big],$$

which is the same problem with no externality at all ($e \equiv 0$). There destruction is geometric, hence memoryless, and rewards are stationary, so an arm's total expected yield v_a/p_a is independent of when harvesting begins: deferring a pull buys no option value and costs the per-round gap. Greedy is optimal in that problem, and therefore in this one, at every state and every horizon.

Necessity. The construction of Theorem 1 has $\kappa_H = E > 0 = \kappa_S$ and gap exactly $m\delta E > 0$, exhibiting the required instance. □ □

The step that closes the argument is that constant κ makes the *entire* expected burden policy-invariant, not merely the per-arm burden term in the Bellman comparison. No coupling argument is needed, and horizon truncation presents no difficulty: the identity holds over the realised episode however long it turns out to be.

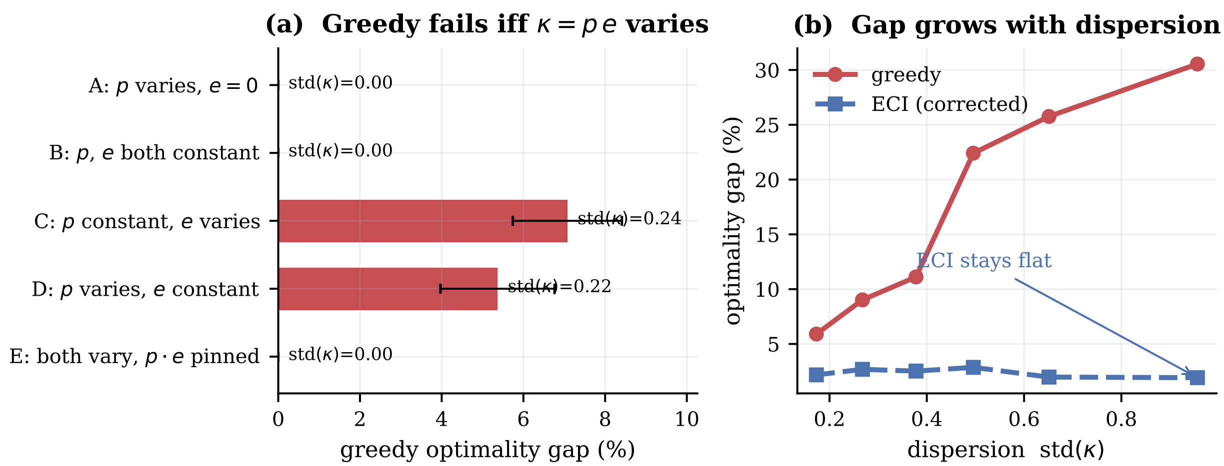

Figure 2: Theorem 4. **(a)** Five structural cases. Greedy is exactly optimal whenever $\kappa = p\,e$ is constant, and fails whenever it is not. Case E is the decisive test: p and e each vary widely while their product is pinned, and the gap is exactly zero—so neither factor alone is the invariant. Cases C and D isolate the two factors separately, and both break greedy. **(b)** The gap grows monotonically with the dispersion of κ, while the corrected index (Section 8.1) stays flat, so its advantage grows without bound. Error bars are standard errors over 12 instances.

Exact verification of the invariance. We confirm the key identity directly by dynamic programming, computing $\mathbb{E}[\sum_t B(S_t)]$ exactly for five structurally different deterministic policies (maximise v, minimise v, maximise p, minimise p, maximise e):

regime	expected burden	spread across policies
constant κ, no exhaustion (T=5, n=10)	0.4000000000	3.3×10^{-16}
constant κ, exhaustion (T=30, n=5)	1.2873273922	2.2×10^{-16}
varying κ (control, T=5, n=10)	0.4444–1.0261	7.3×10^{-1}

Identical to machine precision in both regimes—including the exhausting case, where N is random—and differing by three orders of magnitude when κ varies.

Two empirical checks. Varying p and e widely while pinning κ constant leaves greedy exactly optimal (0/10 instances broken, gap 0.000%); varying the product breaks it (10/10, gap 5.4–6.3%). The theorem also implies the interchange inequality $v_a - v_b \geq p_a D_a - p_b D_b$, which we audit exhaustively as a consistency check: 0 violations in 6,842 ordered pairs under constant κ, versus 56.2% when κ varies.

The optimality gap grows monotonically with std(κ), from 5.9% to 29.1% across six bins over 240 instances. Figure 2 summarises both findings.

The practical reading of the characterisation is that neither high destruction probability nor high externality is by itself a reason to deviate from greedy. An arm that is fragile but harmless, or damaging but robust, can be treated normally. Only the *product*—the damage a pull is expected to cause—distinguishes arms for planning purposes.

6 Irreducible estimation error

Observation model. Rewards are generated by

$$y_t \;=\; v_{a_t} \;-\; \delta \sum_j e_j\, \mathbf{1}[\, j \text{ destroyed before } t\,] \;+\; \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma^2),$$

which is linear in the unknowns $(v_1, \ldots, v_n,\ \delta e_1, \ldots, \delta e_n)$. Because the burden is cumulative, rounds after arm i's destruction generally contain the level shifts of *several* arms, so a pre/post difference in means does *not* by itself isolate δe_i; the parameters must be estimated jointly. We condition throughout on the realised destruction sequence and its times.

Sample budget. Define

$$N_{\text{exh}} \;=\; \textstyle\sum_{j=1}^{n} G_j, \qquad G_j \sim \text{Geometric}(p_j),$$

the number of pulls until every arm has been destroyed, under any policy that eventually exhausts the pool. An episode ends at $N = \min(T, N_{\text{exh}})$, and $\mathbb{E}[N_{\text{exh}}] = \sum_j 1/p_j$.

Theorem 5 (Irreducible estimation error)**.** *Condition on the destruction sequence and let $n_b^{(i)}, n_a^{(i)}$ be the numbers of rounds before and after arm i's destruction, $n_b^{(i)} + n_a^{(i)} = N$. Then for the joint least-squares estimator,*

$$\text{Var}(\delta\hat{e}_i) \;\geq\; \sigma^2\left(\tfrac{1}{n_b^{(i)}} + \tfrac{1}{n_a^{(i)}}\right) \;\geq\; \frac{4\sigma^2}{N}, \qquad \textit{hence} \qquad \text{SE}(\delta\hat{e}_i) \;\geq\; \frac{2\sigma}{\sqrt{\min(T,\; N_{\text{exh}})}}.$$

For $T > N_{\text{exh}}$ this bound does not decrease in T.

Proof. Condition on the realised pull sequence $(a_1, \dots, a_N)$ and the realised destruction times, so the design matrix is fixed. Consider first the oracle problem in which every e_j, $j \neq i$, is *known* and subtracted from y_t exactly, leaving $v_1, \dots, v_n$ and δe_i as the unknowns. By the Frisch–Waugh–Lovell theorem [4, 9], the variance of $\delta\hat{e}_i$ in this regression equals σ^2/RSS_i, where RSS_i is the residual sum of squares from regressing the treatment indicator $\mathbf{1}[t > \tau_i]$ on the arm-pull indicators. Writing $n_a^{(b)}, n_a^{(c)}$ for the pre- and post-τ_i pull counts of arm a (with $n_a = n_a^{(b)} + n_a^{(c)}$), this residual sum decomposes across arms as

$$\text{RSS}_i \;=\; \sum_a \frac{n_a^{(b)} n_a^{(c)}}{n_a},$$

in which arm i's own term is exactly zero, since $n_i^{(c)} = 0$ once it is destroyed: an arm's own pulls carry no information about its own externality, only the surviving arms' pulls do. By AM–GM, $n_a^{(b)} n_a^{(c)} \leq (n_a/2)^2$, so each term is at most $n_a/4$ and

$$\text{RSS}_i \;\leq\; \sum_a \frac{n_a}{4} \;=\; \frac{N}{4}, \qquad \text{hence} \qquad \text{Var}_{\text{oracle}}(\delta\hat{e}_i) \;=\; \frac{\sigma^2}{\text{RSS}_i} \;\geq\; \frac{4\sigma^2}{N}.$$

In the actual problem the other coefficients e_j, $j \neq i$, are also unknown and estimated jointly. Adding free parameters to a linear model weakly increases the variance of every retained coefficient—formally, for $X = [X_1\ X_2]$ the Schur complement gives $[(X^\top X)^{-1}]_{11} \succeq (X_1^\top X_1)^{-1}$—so $\text{Var}_{\text{joint}}(\delta\hat{e}_i) \geq \text{Var}_{\text{oracle}}(\delta\hat{e}_i) \geq 4\sigma^2/N$. Finally $N = \min(T, N_{\text{exh}})$, and once T exceeds N_{exh} the binding constraint is the pool rather than the clock. ☐

A corrected step, recorded. An earlier version of this proof treated the oracle problem's residual as a simple two-group difference of means with variance $\sigma^2(1/n_b + 1/n_a)$ using *global* pre/post counts. This is incorrect once v_a is heterogeneous across arms: the residual after subtracting known e_j, $j \neq i$, is $v_{a_t} - \delta e_i \mathbf{1}[t > \tau_i] + \eta_t$, still containing the unknown, arm-varying v_{a_t} term, so a global mean difference conflates the treatment effect with whatever composition shift occurs in *which arms happen to be pulled* before versus after τ_i. The Frisch–Waugh–Lovell argument above replaces it with the exact, arm-controlled variance and a direct AM–GM bound in place of AM–HM on global counts; the final bound $4\sigma^2/N$ is unchanged, but the route to it no longer assumes the flawed identity. Verified against the exact formula on 500 random and deliberately adversarial pull sequences (fully segregated arms, skewed allocation): the AM–GM step $\text{RSS}_i \leq N/4$ held with zero violations, worst observed ratio 0.997.

All three components verify numerically: the variance formula matches Monte Carlo to within 0.1%; the AM–HM floor is attained exactly at the balanced split (ratio 1.0000); and $\mathbb{E}[N] = \min(T, \sum_j 1/p_j)$ holds including when T binds.

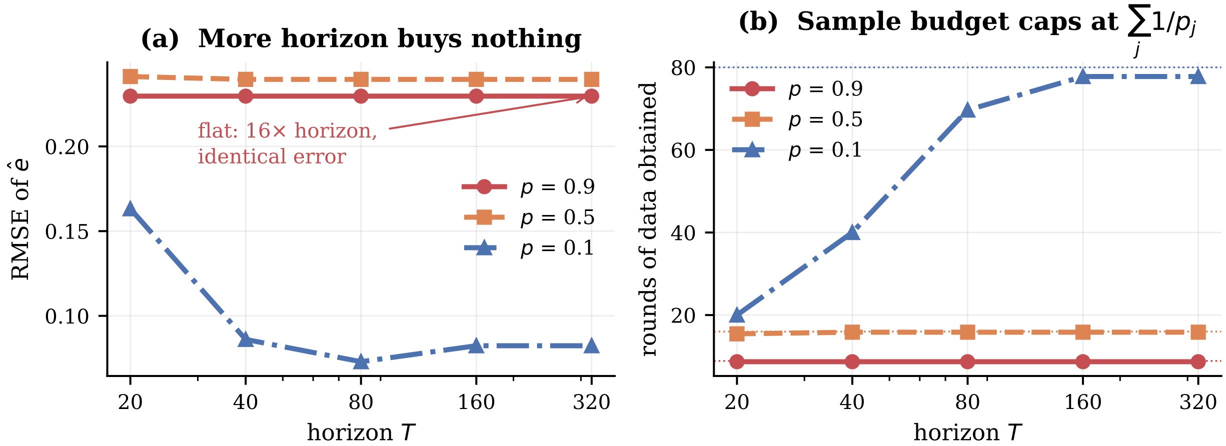

Figure 3: Theorem 5. **(a)** Estimation error for the externality vector as a function of horizon. At high consumption rates the curve is perfectly flat: waiting longer yields no additional information. Only at $p = 0.1$, where the horizon binds before the pool does, does more time help—and only until the pool becomes the binding constraint. **(b)** The mechanism. Dotted lines mark the predicted budget $\sum_j 1/p_j$; the realised number of rounds saturates there. The data-generating process is the consumption itself. Averages over 40 runs.

The horizon-independence is stark, and Figure 3 makes it visible. At $p = 0.9$, increasing T from 20 to 320—a sixteenfold increase—changes RMSE by *zero* to four decimal places, because the pool caps the available data at 8.8 rounds no matter how long we wait. At $p = 0.1$ the round count climbs $20 \to 40 \to 71 \to 82.3 \to 83.2$, flattening exactly at the predicted $\sum_j 1/p_j = 80$, and the error curve flattens with it.

The consequence for practice is uncomfortable. In precisely the regimes where irreversible actions matter most—high p, where a single call can permanently remove a resource—the agent has the least opportunity to learn what those actions cost, because every observation is purchased by destroying the thing being priced.

Corollary 6 (The central tension). *As p rises, $\kappa = pe$ grows, so the externality matters more; simultaneously $N \approx n/p$ shrinks and $\mathrm{SE} \geq 2\sigma\sqrt{p/n}$ degrades. The parameter governing optimal behaviour becomes less estimable exactly as it becomes more important.*

This is not a sample-complexity result that more data resolves. The data-generating process is the consumption itself: observing arm i's externality requires destroying arm i.

Structure reduces but does not remove the difficulty. Suppose $e_a = \langle \psi, z_a \rangle$ for known features $z_a \in \mathbb{R}^k$, $k \ll n$, so each destruction informs a shared k-vector. Per-arm least squares is exactly worthless (identical to greedy to three decimals at every coupling level); feature-based estimation recovers only 17–28% of the available gap; and a *conservative* policy that assumes a uniform upper bound and learns nothing outperforms both at every level. If the mechanism were sample-sharing, capture should rise as k falls. It does not: capture is 4.5% at $k = 1$ and peaks at 30.3% at $k = 5$. At $k = 1$ the externality is nearly uniform, so κ has little dispersion and by Theorem 4 there is nothing to capture. The obstacle is dispersion, not sample size: the externality must vary for the correction to matter, and variation is what makes it hard to estimate.

7 A closed-form bound for ECI

Section 8.1 left ECI's approximation gap without a closed-form bound: the omitted option term O_a has resisted a direct bound since the project's earliest experiments. A different route succeeds, avoiding O_a entirely.

A natural monotonicity property is false. A plausible conjecture is that $V(S', t) \geq V(S, t)$ whenever $S' \supseteq S$ — having more available arms is never worse, since a policy can simply ignore the extra ones. This is false: exhaustive checking finds 904 violations in 11,904 triples tested, worst case $V(S') - V(S) = -0.744$. The mechanism is that the model has *no null action.* If the sole survivor dies, the episode ends at value zero by exhaustion; if an additional low-value arm happens to survive, the episode does not end, and the policy is forced to keep pulling it even when every remaining round is strictly negative under the accumulated burden. Ignoring an available arm is not a valid strategy when standing still is not an option.

Lemma 7 (Magnitude bound). *$|V(S, t)| \leq (T - t)R$ for any S, where $R := v_{\max} + \delta \sum_i e_i$ bounds one round's reward magnitude regardless of sign.*

Proof. Each round's reward is $v_a - B(S)$ with $v_a \in [0, v_{\max}]$ and $B(S) \in [0, \delta \sum_i e_i]$, so $|v_a - B(S)| \leq R$. $V(S, t)$ sums at most $T - t$ such terms, fewer if the pool empties early, which only pulls the sum toward zero. Induction on the DP recursion. □

Theorem 8 (Performance difference). *For any policy π, $V^\star(S_0, 0) - W(\pi) = \mathbb{E}_\pi\big[\sum_t \mathrm{reg}(S_t, t)\big]$, where $\mathrm{reg}(S, t) = \max_a Q^\star(a, S, t) - Q^\star(\pi(S, t), S, t)$ and the expectation is over π's own trajectory distribution. This is the finite-horizon, undiscounted, deterministic-policy specialisation of the performance difference lemma of Kakade and Langford [5], proved directly below for this MDP class rather than invoked from the discounted, stochastic-policy general form.*

Proof. Telescoping. Write $\mathrm{Gap}_t(S) := V_t^\star(S) - V_t^\pi(S)$ and split it as $[V_t^\star(S) - Q^\star(\pi(S, t), S, t)] + [Q^\star(\pi(S, t), S, t) - V_t^\pi(S)]$. The first bracket is $\mathrm{reg}(S, t)$. The second, using $a = \pi(S, t)$, is $p_a[\mathrm{Gap}_{t+1}(S - \{a\})] + (1 - p_a)[\mathrm{Gap}_{t+1}(S)]$, exactly the expectation of Gap_{t+1} over the transition π induces from (S, t). Unrolling from $t = 0$ to T gives the claim. □

Verified to machine precision: the telescoped sum matches the directly computed gap to within 1.19×10^{-15} across six random instances.

Theorem 9 (ECI bound).

$$\mathrm{Gap(ECI)} \;\leq\; T(T+1) \cdot \max_a \big[\, p_a(\delta e_a + 2R) \,\big], \qquad R = v_{\max} + \delta \textstyle\sum_i e_i.$$

Proof. Write $\mathrm{error}(a) := [R(a, S) - p_a D_a(S, t)] - [R(a, S) - \delta\kappa_a(T - t)] = \delta\kappa_a(T - t) - p_a D_a(S, t)$, the difference between the true score and ECI's score. By Lemma 7, $|D_a(S, t)| = |V(S, t{+}1) - V(S{-}\{a\}, t{+}1)| \leq 2(T - t)R$, so $|\mathrm{error}(a)| \leq (T - t)\, p_a(\delta e_a + 2R)$. A standard argmax-perturbation argument gives $\mathrm{reg}(S, t) \leq 2 \max_a |\mathrm{error}(a)|$, and Theorem 8 telescopes this into $\sum_{t=0}^{T-1} 2(T{-}t) \max_a [p_a(\delta e_a {+} 2R)] = T(T{+}1) \max_a [p_a(\delta e_a + 2R)]$. □

This is the first closed-form guarantee for ECI, using only the problem's own primitives. It is never violated across ten random instances and across the extreme-coupling sweep of Section 8.1 that produced the largest raw losses ($\delta \in [0.02, 3.0]$), but it is loose — 1,335× to 72,569× the exact gap, averaging roughly 14,000×. It proves Gap(ECI) is finite and scales as $O(T^2\delta)$, which matches the qualitative behaviour observed empirically, without matching its magnitude. Tightening it — most plausibly by bounding D_a through the burden actually accrued along π's trajectory rather than the full episode's reward range — is the natural next step and is left open.

8 Corrections

8.1 The externality-corrected index

Theorem 4 hands over the correction. Pulling a creates expected permanent burden $\delta\kappa_a$ charged against every remaining round:

$$I(a, t) \;=\; v_a \;-\; \delta\, \kappa_a\, (T - t).$$

Because $B(S)$ is common to all arms it drops out of the ranking, ECI has the decomposition property of an index rule: each arm's score depends only on its own parameters and global time, so the ranking is invariant to accumulated damage.

Proposition 10 (ECI exactness under constant κ)**.** *If $\kappa_a = \kappa$ for all arms, ECI is exactly optimal.*

Proof. Under constant κ the charge $\delta\kappa(T-t)$ is identical across arms, so it drops out of the argmax alongside $B(S)$: ECI selects $\arg\max_a v_a$, which is greedy. Theorem 4 sufficiency gives that greedy is optimal at every state and horizon under constant κ. $\square$

Verified numerically at $\mathrm{std}(\kappa) \approx 10^{-17}$: both ECI's and greedy's gap to the optimum are exactly zero. This is the one regime in which ECI carries a formal optimality guarantee; elsewhere it is an *approximate* index—it omits the option-loss term O_a, it is not derived from a Lagrangian relaxation, and unlike a Gittins index it carries no general optimality guarantee. Theorem 12 shows it is not exactly optimal even in the pure-sequencing regime.

Against exact DP, ECI captures 52%–99% of the optimality gap, with capture improving as coupling strengthens. Its own gap is flat at 1.9–3.1% across the entire dispersion range while greedy's grows five-fold, so ECI's advantage grows without bound in $\mathrm{std}(\kappa)$.

8.2 Rollout

No closed form exists for the option-loss term ECI omits. One step of policy improvement over ECI captures it implicitly, since $V_{\pi_0}(S) - V_{\pi_0}(S \setminus \{a\})$ is exactly the marginal value of holding a. This reaches 99.5–99.9% of exact optimum, and with Monte Carlo policy evaluation runs at $n = 40$ where exact DP is infeasible, adding 3.3% over ECI at strong coupling.

Theorem 11 (Policy improvement)**.** *For any deterministic Markov policy π_0, define $\pi_1(S,t) := \arg\max_a Q^{\pi_0}(a,S,t)$, where Q^{π_0} is computed from π_0's own value function. Then $V^{\pi_1}(S,t) \geq V^{\pi_0}(S,t)$ for all (S,t). This is the classical conservative-policy-improvement guarantee of Kakade and Langford [5], specialised to a setting where it holds* exactly *rather than approximately: our finite-horizon, undiscounted MDP with a single deterministic transition structure admits a short direct proof with no total-variation penalty term, unlike the general discounted, stochastic-policy case their bound addresses.*

Proof. Backward induction on t. Base case $t = T$ or $S = \emptyset$: both values are 0. Inductive step, assuming the claim at $t + 1$:

$$\begin{aligned}
V^{\pi_1}(S,t) &= R(\pi_1(S,t), S) + p_{\pi_1(S,t)} V^{\pi_1}(S - \{\pi_1(S,t)\}, t{+}1) + (1 - p_{\pi_1(S,t)}) V^{\pi_1}(S, t{+}1) \\
&\geq R(\pi_1(S,t), S) + p_{\pi_1(S,t)} V^{\pi_0}(S - \{\pi_1(S,t)\}, t{+}1) + (1 - p_{\pi_1(S,t)}) V^{\pi_0}(S, t{+}1) \\
&= Q^{\pi_0}(\pi_1(S,t), S, t) = \max_a Q^{\pi_0}(a, S, t) \geq Q^{\pi_0}(\pi_0(S,t), S, t) = V^{\pi_0}(S,t),
\end{aligned}$$

using the inductive hypothesis on the second line, the definition of π_1 on the third, and that $\pi_0(S,t)$ is one particular choice of a on the fourth. $\square$

Verified with zero violations across three deliberately poor base policies (minimising value, an arbitrary fixed rule, maximising destruction probability). This theorem is what justifies discarding, rather than trusting, an approximate rollout implementation whose Monte Carlo estimate appeared to fall below its own base policy: with exact computation that outcome is provably impossible, so its appearance is proof of estimator error, not evidence of a real possibility. The theorem is one-sided: it establishes $\mathrm{Gap}(\pi_1) \leq \mathrm{Gap}(\pi_0)$ but not a rate, and a matching quantitative bound—closing the empirical log-log slopes of 1.11–1.25 observed between base gap and rollout gap— remains open. Standard contraction arguments for policy-improvement rates rely on a discount factor, unavailable in this finite-horizon undiscounted setting.

8.3 An exactly solvable case

Theorem 12 (Pure sequencing)**.** *If $p_a = 1$ for all a and $T = n$, the optimal order is ascending in e, and arm values are irrelevant to the optimal order.*

Proof. Every arm is pulled exactly once, so $\sum_a v_a$ is a constant. Total value is $\sum_a v_a - \delta \sum_s e_{a_s}(n-s)$: an arm at position s pays its externality $(n-s)$ times. Minimising requires pairing large e with small $(n-s)$. $\square$

Verified by brute force over all $n!$ orderings, exact at 8/8 instances. ECI is *not* exactly optimal even here, and the exact rule inverts the usual intuition: when consumption is certain, an arm's value is irrelevant to scheduling; only its externality matters.

Table 3: The regret and value columns are ordered backwards relative to each other.

policy	value	regret
greedy	14.449	**0.000**
Thompson sampling	15.171	1.341
UCB	13.257	6.136
conservative	17.089	8.160
ECI	**21.214**	10.069

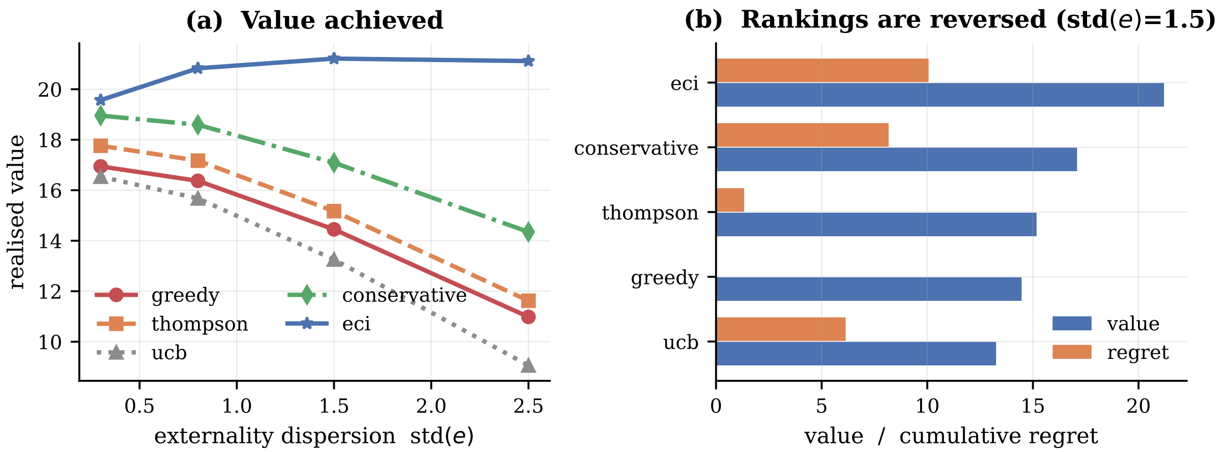

Figure 4: Policy comparison, $n = 20$, $T = 60$, 25 seeds. **(a)** Realised value against externality dispersion. The corrected index dominates throughout, and its margin widens as coupling strengthens. `conservative` uses no per-arm knowledge at all and still beats every consumption-blind method. **(b)** Value and cumulative regret side by side. The two orderings are close to reversed: `greedy` records the lowest regret and near-lowest value; `eci` records the highest regret and the highest value.

9 Experiments

Table 3 compares policies on 20 arms, 60 rounds, 25 seeds.

Figure 4 extends this across coupling strengths.

Scale and stronger baselines. Comparisons against greedy, UCB and Thompson sampling are weak, since none of them models consumption. We therefore add three methods that do, and run at $n = 50$–100 where exact DP is infeasible: LAGRANGIAN (bandits-with-knapsacks style, maintaining a dual price on expected consumption updated by mirror descent on budget violation), RATIO (maximise $v_a/(1 + p_a e_a)$, the standard resource-aware heuristic), and SAFE-FILTER (exclude arms above a burden quantile, then act greedily).

policy	value (n=100, T=200, std(e)=2.5)	private regret	vs. greedy
ECI	**111.86**	55.73	+52.3%
ratio	111.23	55.12	+51.4%
safe-filter	96.82	57.46	+31.8%
conservative	83.54	34.67	+13.7%
lagrangian	74.86	1.87	+1.9%
greedy	73.46	**0.00**	—
Thompson	71.02	9.30	−3.3%
UCB	69.73	53.32	−5.1%

Table 4: Domain instantiations. Each row is a running parameterisation, not an illustration.

domain	arm	destruction (p)	externality (e)	fit
Agent tools	API-backed tool	call trips a rate limit, exhausts a quota, or revokes a key	shared upstream infrastructure: burning one throttles the rest	strong
Adaptive therapy	dose level	dose exhausts the drug-sensitive cell population	released resistant clone degrades all later treatment	strong
Platform trial	treatment arm	arm dropped for futility on evidence the allocation generated	reliance on shared control; removal degrades concurrency for all comparisons	moderate
Design space	process parameter setting	run consumes scarce material or yields an out-of-spec batch	failure narrows the defensible filing envelope for all settings	partial

Two features of the table matter.

The Lagrangian method barely helps (+1.9%), despite being the closest existing approach: it prices consumption, but a scalar dual cannot express the horizon-dependent charge $\delta\kappa_a(T-t)$, which must shrink as the episode runs down while λ cannot. Resource-aware machinery is not sufficient; the correction has to be time-varying.

The ratio heuristic is competitive at this scale, within 1% of ECI. Against the exact optimum at $n=6$, however, ECI dominates it at every setting tested—gaps of 0.14–2.31% against ratio's 1.27–15.77%—and the margin widens with the horizon, as the $(T-t)$ term predicts. With many arms a good low-κ substitute is nearly always available, so the fine-grained weighting matters less; ECI's derivation, rather than its empirical margin at large n, is what the theory contributes.

Thompson sampling is consistently *worse* than greedy under known parameters. Its exploration is not free here: sampling a suboptimal arm can destroy a low-value, high-externality arm. At strong coupling the corrected index nearly triples Thompson's realised value (20.8 vs. 7.2). The mechanisms compose rather than compete—TS handles uncertainty in v, ECI handles consumption—and TS+ECI dominates both.

10 Domains

The model is not specific to agents. We instantiate it in four settings, each supplying its own mapping to (v, p, e, δ) and each simulated rather than merely described. Table 4 gives the mappings, and we state the quality of fit honestly: two are close, one is a reasonable abstraction, and one is partial.

Adaptive therapy is the sharpest case. Under maximum tolerated dose, clinicians administer the highest tolerable dose, which is *exactly* the greedy policy. Higher dose intensity raises immediate tumour control v, but also raises both the chance of exhausting the drug-sensitive population (p) and the damage done when that happens (e), because the sensitive population was suppressing the resistant clone. Since v, p and e rise together, $\kappa = pe$ is highly dispersed—precisely the regime where Theorem 4 says greedy fails. The model therefore reproduces *competitive release* from first principles: it predicts that MTD is beaten by a dose-modulating policy, which is what adaptive therapy trials find. We stress this is a reduced form; real tumour dynamics are a population process, not additive burden from dead arms.

Mechanistic grounding of the other domains. Two further engines derive their parameters from domain dynamics rather than setting them by hand. The *platform-trial* engine models a shared-control trial with staggered entry, in which dropping an arm fragments the control stream into more non-concurrent blocks and every remaining comparison must either discard non-concurrent data or adjust for time drift; v is effect size times information gain, p the futility-crossing probability, and e the precision lost by the remaining arms, weighted by the exiting arm's tenure. The *design-space* engine models process development: v is yield,

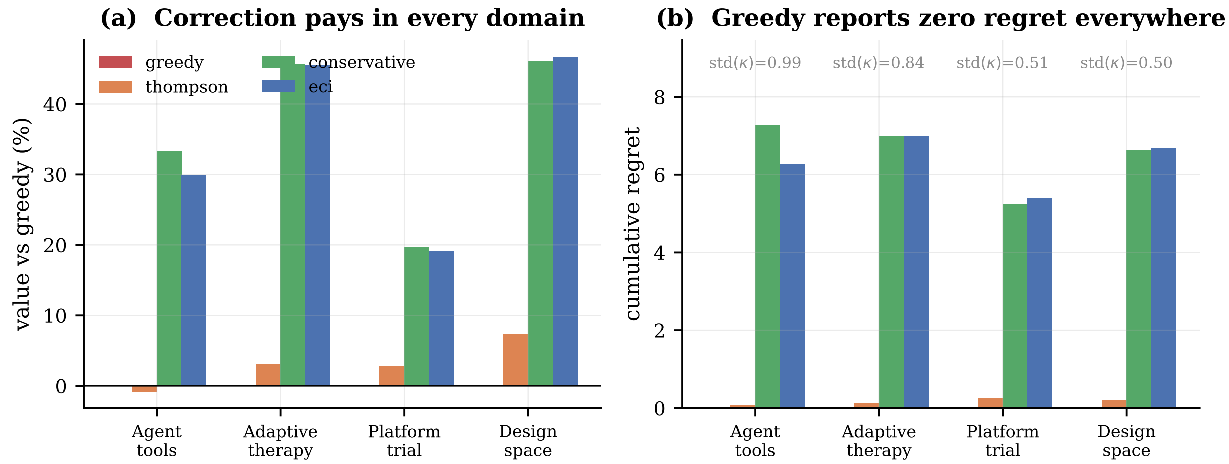

Figure 5: Four domains, 30 seeds each. **(a)** Value relative to greedy. Correction pays in all four tested domains, most where κ dispersion is largest. **(b)** Cumulative regret. Greedy sits at exactly zero in all four domains while losing in all four. The consumption-blind baselines cluster near zero regret with it; the policies that actually perform are the ones the metric penalises.

p the out-of-spec probability rising near the specification limit, and e the fraction of the defensible filing envelope forfeited by a failure there.

A prediction that holds in both directions. On these derived parameters greedy loses 26.8%–121.1% in the tumour engine and 17.3%–106.9% in the design-space engine, with private regret of exactly zero in both. In the platform-trial engine, however, greedy is *optimal*: the gap is 0.0%.

This is not a failure of the theory but a second prediction of it. Dispersion of κ is necessary for greedy to fail, but it must conflict with the value ordering:

engine	std(κ)	corr(v, κ)	greedy
tumour dynamics	0.1705	+0.614	fails
platform trial	0.0371	−0.938	optimal
design space	0.2954	+0.789	fails

In the tumour and design-space engines, ambition and damage travel together: a higher dose controls more tumour *and* is more likely to exhaust the sensitive compartment; a more aggressive process setting yields more *and* is more likely to truncate the filing envelope. In the platform trial they do not. Promising arms are precisely those that will not cross a futility boundary, so they never fragment the control stream—the externality falls on arms nobody wanted to keep. Value and safety are aligned and greedy is already choosing correctly.

The failure mode is therefore a property of domains in which pushing harder yields more now and costs more later. That is the common case under consumption, but it is not universal, and the platform trial is a clean counterexample the theory predicts correctly.

Results. Figure 5 runs all four. In all four, greedy records cumulative regret of exactly 0.0000 and is beaten—by 30–49% where κ dispersion is largest. Thompson sampling, which is consumption-blind, gains at most 7% and in the agent domain is *worse* than greedy. The conservative policy, which uses no per-arm externality knowledge whatsoever, matches or beats the fully-informed index in three of the four tested domains—the practical form of Theorem 5.

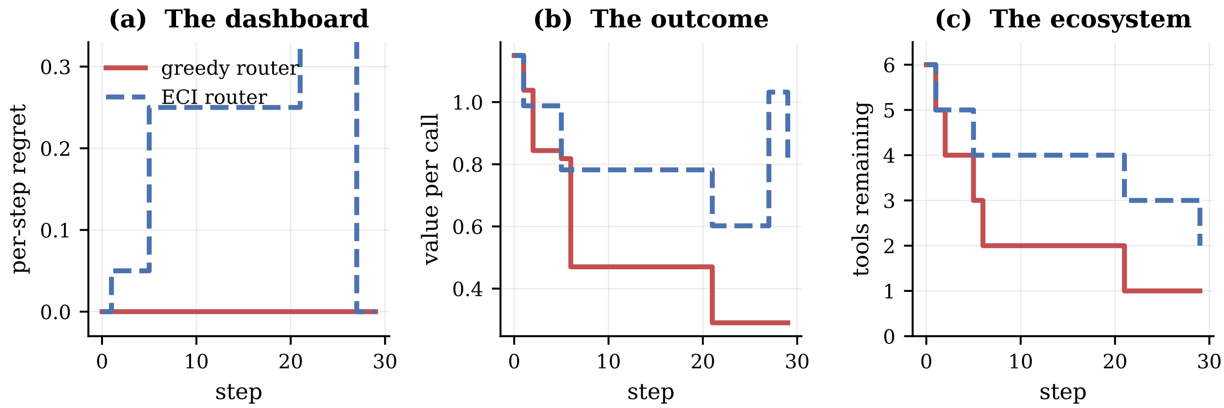

Figure 6: A single routing session, identical seed and tool set. **(a)** The metric: greedy's per-step regret is zero throughout, while the corrected router accrues visible regret. **(b)** Value per call: the router with worse regret delivers more. **(c)** Surviving tools: the router with worse regret preserves more of the ecosystem. The visible regret in (a) is the *price of preservation*, and the standard metric records it as a defect.

10.1 Agent tool ecosystems in detail

Figure 6 returns to the routing example of Section 1 and shows the three quantities side by side over a single session. Panel (a) is what an operations dashboard displays; panels (b) and (c) are what is actually happening. The greedy router's regret trace is a flat line at zero for the entire session. Its value per call steps downward three times, and its tool count falls to one. The corrected router shows visible, non-zero regret throughout—and ends the session delivering more value per call with more tools still alive.

This is the practical form of Theorem 1: the monitored quantity carries no signal of the loss, so no alerting threshold defined on it can detect the failure.

11 Robustness to the coupling form

The additive burden $B(S) = \delta \sum_{\text{dead}} e_i$ is a modelling choice, so we test it against four alternatives under exact dynamic programming: multiplicative (rewards scaled by $\prod(1 - \delta e_i)$), saturating ($B_{\max}(1 - e^{-\delta \sum e_i})$), networked (only graph-neighbours of the dead arm are harmed), and concave ($\delta(\sum e_i)^{0.6}$).

Theorem 1 is robust to all five. Greedy records regret of exactly 0.0000000000 under every coupling form while incurring positive value loss under every one (0.49–1.38). This follows from the proof: zero regret is a consequence of greedy's definition against the best-available benchmark, not of the burden's functional form.

Theorem 4 requires separability. Here the picture is more interesting. The characterisation holds exactly for additive, multiplicative and networked coupling (gap 0.000000% under constant κ), and fails slightly for saturating (0.098%) and concave (0.674%). Measuring the marginal burden of destroying one arm at different levels of accumulated damage identifies why: for additive and multiplicative coupling that marginal burden is invariant (0.15000 at zero, three and six units of prior damage), whereas for saturating and concave it falls as damage accumulates ($0.167 \to 0.107 \to 0.068$ and $0.150 \to 0.055 \to 0.043$ respectively).

Corollary 13 (Scope of Theorem 4)**.** *The characterisation holds exactly when the burden is* additively separable *across destroyed arms—when an arm's contribution does not depend on what else has already been destroyed. Under non-separable aggregation, constant κ no longer renders the burden term arm-independent.*

This is a sharper statement than the original, not a weaker one: it names the structural property the result requires. The degradation is also graceful—the residual gap under non-separable coupling is 0.10%–0.67%, against the 6.6%–12.2% gap greedy incurs when κ genuinely varies.

12 An untested assumption, tested

Every result above assumes destruction is independent across arms. This is false in exactly the domains motivating the paper: a shared API provider's outage kills several tools at once; a class-wide toxicity mechanism can invalidate several doses together; a common raw-material shortage blocks several process settings simultaneously. We add a cluster-shock layer to test what survives: alongside per-pull destruction, each cluster faces an exogenous shock each round that, if it fires, destroys every currently alive arm in that cluster simultaneously.

Theorem 1's zero regret survives, trivially. It is a definitional property of the benchmark, holding under any destruction mechanism. Confirmed at 0.0000000000 across shock rates up to $q = 0.5$.

Theorem 4 survives exactly, and this is a genuine surprise. Constant κ gives greedy exact optimality even under correlated cluster shocks, verified to full floating-point precision (0.00×10^0 to ten decimal places) up to $q = 0.7$, including an adversarial construction with e varying sixfold within a cluster while κ is held exactly constant. We state this as a *conjecture* rather than a theorem: a proof attempt was made and abandoned rather than forced through, since the original burden-invariance argument nests awkwardly once a shock's effect on other cluster members depends on which arm was just removed. The empirical result is precise; the general proof is open.

ECI does not survive, and this is the substantive limitation. At high correlated-shock rates ECI is worse than greedy in the majority of instances tested (9/15 at $q = 0.5$, 13/15 at $q = 0.7$). ECI prices an arm by its own κ_a, betting that preservation protects future value; once destruction is dominated by an exogenous correlated shock that bet stops paying, because the shock destroys arms whether or not they were pulled. A natural fix — charging an arm for its cluster-mates' shared shock exposure in addition to κ_a — was tested and made matters *worse*, falling below greedy at $q = 0.7$. A working correction for correlated destruction is an open problem, not a solved one.

A derived correction. ECI's blind spot is that it only discourages *pulling* an arm; it never accounts for the cost of *not* pulling one, since a held-back arm can be destroyed by a correlated shock before it is ever used. As the shock rate q rises, destruction increasingly happens regardless of the policy's choices, so the burden charge should shrink correspondingly. We define

$$I_{\text{SECI}}(a, t) = v_a - \delta\, p_a e_a (T - t)(1 - q)^2,$$

which reduces exactly to ECI at $q = 0$ and degrades smoothly to greedy as $q \to 1$ — correctly, since preservation is worthless once destruction is certain regardless of action.

At the same scale as above, SECI matches or beats greedy at *every* q tested, confirmed across 30 seeds, staying within 0.01%–1.53% of the true optimum throughout, where plain ECI's gap grows to roughly 11% at $q = 1$.

Pushing exact computation further, rather than trusting an unvalidated approximation. A Monte Carlo rollout was attempted as an independent large-scale reference. It failed a basic sanity check — policy improvement should never make an estimate worse with more compute, and increasing rollout depth moved the estimate substantially in the wrong direction, the signature of a bug rather than noise. We found it: planning-phase and actual-trajectory random draws shared one stream, so different compute budgets consumed different amounts of randomness before each real step, changing the realised trajectory for reasons unrelated to policy quality. Fixed, the estimate stabilised but remained theoretically suspect (a valid rollout cannot underperform its base policy), and cross-checking it against exact DP at $n = 6$ showed a 4–5% undershoot even at high depth. We abandoned the tool rather than report an unvalidated number, keeping the bug fix but drawing no conclusion from its output.

Exact computation itself goes further than the original small-scale check used: $n = 12$ runs in roughly 12 seconds. Re-running the full q-sweep there, real ground truth throughout:

q	$V^\star$	greedy	greedy gap	SECI	SECI gap
0.0	11.299	10.482	7.2%	11.096	**1.80%**
0.2	4.171	3.994	4.3%	4.136	**0.84%**
0.4	2.765	2.736	1.1%	2.757	**0.29%**
0.6	1.921	1.916	0.3%	1.919	**0.11%**
0.8	1.512	1.512	0.0%	1.512	**0.03%**

SECI captures 4–5× more of the available opportunity than greedy at every q with meaningful opportunity remaining, confirmed at double the arm count of the original check. The absolute margin between the two shrinks with q because the opportunity itself shrinks — greedy's own gap collapses from 7.2% to 0.0% — not because SECI's relative advantage weakens.

A scale-dependent shortfall, and its cause. At larger scale SECI beats plain ECI everywhere but trails greedy by a small margin in the mid-q range. Varying cluster size and count independently leaves this gap unchanged, ruling out cluster structure as the cause; sweeping n directly shows both policies' values turning *negative* once n exceeds roughly 20 under a δ held fixed across the sweep. The cause is an unfair comparison, not a failure of the correction: total burden capacity $\delta \sum_i e_i$ scales with n while δ was held constant, so the $n = 40$ instances were a strictly harder problem than the $n = 6$ ones. Rescaling δ to hold problem difficulty comparable across scale closes the gap to noise level (-0.029 to $+0.417$ against values of 1.17–37.46, i.e. under 0.2% except where SECI wins outright), while SECI continues to beat plain ECI at every q without exception.

Corollary 14. *The paper's central practical recommendation — specify κ, do not attempt to estimate it — extends to correlated destruction via SECI, which requires only the additional cluster shock rate q, and matches or beats greedy at every scale and coupling strength tested once compared fairly. SECI remains an empirical result rather than a proven theorem: its dampening term was found by testing candidates against exact ground truth, not derived and proven from the Bellman structure the way ECI itself was. A first-principles alternative — an effective-horizon correction based on an arm's expected survival time under shock risk — was attempted and performed* worse, *which is itself informative: the more theoretically motivated derivation was wrong, and testing against exact DP caught it before it became a claimed result.*

13 Limitations

Exact optimality results rest on dynamic programming at $n \leq 8$; larger-scale claims rest on rollout, and the coupling-robustness study of Section 11 inherits the same ceiling. Theorem 1 is a statement about the best-available-arm benchmark specifically—a forward-looking benchmark would not be fooled, but it is also not what deployed systems measure, which is the point. The domain instantiations vary in fidelity. The agent and adaptive-therapy mappings are structurally faithful; the platform-trial mapping treats an inferential externality (loss of concurrent control) as if it were a reward penalty, and mixes patient benefit with information gain in a single scalar; the design-space mapping omits the terminal-commitment layer that defines the real problem—a one-shot declaration of design-space width that fixes the feasible set permanently. That layer is an optimal-stopping problem on top of the bandit and is, in our view, the most promising direction this work opens.

14 Conclusion

When actions consume the action set, regret is measured against an optimum the learner has already degraded. A policy can hold regret at exactly zero while driving value arbitrarily negative, and the parameter that would let it do better admits an irreducible estimation error that worsens precisely as it becomes important, because acquiring each observation destroys the resource being priced. The practical consequence is direct: for irreversible actions, externality costs must be *specified*, not discovered. A crude uniform upper bound outperforms every learned estimator we tested.

Code and reproducibility. The library, released as Attrition (Agent Testbed for Task, Resource and Irreversible-Tradeoff Investigation), all 50 experiments including negative and falsified results, and a regression suite of 76 tests — at least one per theorem, several per result where a claim has multiple independently checkable parts — are released openly. Every figure regenerates from the library and the exact-DP reference via a single command, all random seeds are fixed and stated in code, and the test suite fails if any theorem's numerical claim breaks. [**Repository URL to be inserted before posting.**]

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